



# Autonomous Cognitive Agent Flying Aircraft in X-Plane

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# Background

## X-Plane

- Physics-based flight simulation platform
- Realistic environment for autonomous AAM and custom flight operations
- Test environment for cognitive agent's decisions

## SOAR Agent

- Cognitive architecture that models human-like decision making
- Rule-based reasoning to evaluate situations
- Receives information from X-Plane and determines how the aircraft should respond



# Advisor / Client

**Dr. Siddhartha Bhattacharyya**

- Department of Electrical Engineering and Computer Science
- Program Chair Computer Science and Software Engineering

## Meetings

- August 27 @ 10 a.m.
- August 28 @ 2 p.m.





# Goals & Motivation

## Goals

- Extend the existing X-Plane and SOAR framework
- Develop a cognitive agent that can adapt the flight plan in response to aircraft equipment failures

## Motivation

- Create a safe and adaptable autonomous flight system
- Current models are adapted to specific problems with static flight plans



# Key Features

## The user can...

- Define the original departure and destination airports for the cognitive agent's flight route
- Create and simulate an aircraft equipment failure
- Create weather and human-population density scenarios
- Observe the flight route change in response to new conditions



# Novel Features

- **Equipment failure handling**
  - Extends the system, beyond normal flight operations by allowing it to operate under degraded aircraft conditions
- **Dynamic flight routes**
  - The planning flight route can change with new conditions



# Algorithms & Tools

- **SOAR Production Rules / SOAR Architecture**
  - Agent response and memory updates
- **IntelliJ**
  - Java integrated development environment
- **Java**
  - Primary programming language
- **X-Plane Datarefs**
  - X-Plane flight data access and control
- **Maven**
  - Java project configuration and dependency-management tool
- **XPC Plugin (X-Plane Control)**
  - Software control and communication with X-Plane



# Technical Challenges

- **SOAR architecture**
  - New cognitive architecture
  - Learn production rule system, memory model, and development workflow
- **X-Plane & Aircraft Behavior**
  - Aircraft systems, controls, physics, and engineering behavior
  - Model failures and interpret aircraft's response
- **Existing Codebase**
  - Understand current implementation and communication among SOAR, Java, and X-Plane
  - Extend the code without breaking the existing framework

1. **Compare & Select Technical Tools for SOAR, X-Plane, and Secant**
  - SOAR/X-Plane Integration Demo
2. **Resolve initial technical challenges**
  - Soar architecture
  - X-Plane and aircraft behavior
  - Existing codebase
3. **Compare and select collaboration tools**
  - GitHub
  - Google Drive
  - Group chat & Email
  - Jira
4. **Create Requirement Document**
5. **Create Design Document**
6. **Create Test Plan**



# Milestone 1

1. *Implement, test, and demo* - Triggering and communicating equipment failure from X-Plane to the SOAR agent
2. *Implement, test, and demo* - Agent following a complete flight plan in X-Plane with NO complications
3. *Implement, test, and demo* - Agent recognizes and responds to equipment failure
4. *Implement, test, and demo* - Agent issues appropriate commands to the aircraft



## Milestone 2

1. ***Implement, test, and demo*** - Failure generation system with multiple component failures
2. ***Implement, test, and demo*** - Multiple failures and solution visualization on the GUI
3. ***Implement, test, and demo*** - Cognitive agent response to multiple component failures in parallel
4. ***Implement, test, and demo*** - Complete flight plan with one or more equipment failures



## Milestone 3

# Task Matrix

| Task                                 | Dylan                     | Zach     | Avery         | Brianna       |
|--------------------------------------|---------------------------|----------|---------------|---------------|
| Compare and Select Technical Tools   | 25%                       | 25%      | 25%           | 25%           |
| SOAR/X-Plane Integration Demo        | 30%                       | 30%      | 20%           | 20%           |
| Resolve Technical Challenges         | 25%                       | 25%      | 25%           | 25%           |
| Compare & Select Collaboration Tools | Documents / Presentations | Software | Communication | Task Calendar |
| Requirement Document                 | 20%                       | 20%      | 20%           | 40%           |
| Design Document                      | 20%                       | 40%      | 20%           | 20%           |
| Test Plan                            | 40%                       | 20%      | 20%           | 20%           |

**Any Questions?**

